PHYS4055 Week 7: 10/9/26:

1. Separate the schrödinger equation into separate radial & angular parts $(P(r)/r)$ $(Y(\theta,\phi))$

$$\left(\frac{p^2}{2m} + V(r)\right)\psi = E\psi \quad , \quad \left(\frac{-\hbar^2\nabla^2}{2m} + V(r)\right)\frac{P(r)}{r}Y(\theta,\phi) = E\frac{P(r)}{r}Y(\theta,\phi)$$

(H; $p = -i\hbar\nabla$)



Ansatz: $\psi(r,\theta,\phi) = \frac{P(r)}{r}Y(\theta,\phi)$

$\sim \nabla^2\left(\frac{P(r)}{r}Y(\theta,\phi)\right)$ ①

Recall: $\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}$ (spher. coords)

$$① = \nabla^2\left(\frac{P(r)}{r}\right)Y(\theta,\phi) + \frac{P(r)}{r}\nabla^2 Y(\theta,\phi)$$

(A) (B)

$$(A) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\left(\frac{P(r)}{r}\right)\right)Y(\theta,\phi)$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\left[\frac{\partial P(r)}{\partial r}\frac{1}{r} - \frac{1}{r^2}P(r)\right]\right)Y(\theta,\phi)$$

$$= \frac{1}{r^2}\left(\frac{\partial^2 P(r)}{\partial r^2}r + \cancel{\frac{\partial P(r)}{\partial r}} - \cancel{\frac{\partial P(r)}{\partial r}}\right)Y(\theta,\phi)$$

$$= \frac{1}{r}\frac{\partial^2 P(r)}{\partial r^2}Y(\theta,\phi)$$

~~Notes~~

$$\nabla^2 Y(\theta,\phi) = \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y(\theta,\phi)}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 Y(\theta,\phi)}{\partial\phi^2}$$

$$= \frac{1}{r^2\sin\theta}\left[\cos\theta\frac{\partial Y(\theta,\phi)}{\partial\theta} + \sin\theta\frac{\partial^2 Y(\theta,\phi)}{\partial\theta^2}\right] + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}Y(\theta,\phi)$$

"

$$(B) = \frac{P(r)}{r}$$ "

Original equation:

$$-\frac{\hbar^2}{2m}\nabla^2\frac{P(r)}{r}Y(\theta,\phi) = (E - V(r))\frac{P(r)}{r}Y(\theta,\phi)$$

LHS:

$$= -\frac{\hbar^2}{2m}\left[\frac{1}{r}\frac{\partial P}{\partial r^2}Y + \frac{P}{r}\left(\frac{1}{r^2\sin\theta}\left\{\cos\theta\frac{\partial Y}{\partial\theta} + \sin\theta\frac{\partial^2 Y}{\partial\theta^2}\right\} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right)\right]$$

RHS:

$$= (E - V(r))\frac{P}{r}Y$$

Rearranging,

$$(E - V(r)) = -\frac{\hbar^2}{2m}\left[\frac{1}{P}\frac{\partial^2 P}{\partial r^2} + \frac{1}{r^2\sin\theta Y}\left\{\cos\theta\frac{\partial Y}{\partial\theta} + \sin\theta\frac{\partial^2 Y}{\partial\theta^2}\right\} + \frac{1}{r^2\sin^2\theta Y}\frac{\partial^2 Y}{\partial\phi^2}\right]$$

$$r^2(E - V(r)) + \frac{\hbar^2}{2m}\left(\frac{r^2}{P}\frac{\partial^2 P}{\partial r^2}\right) = \frac{1}{\sin\theta Y}\left\{\cos\theta\frac{\partial Y}{\partial\theta} + \sin\theta\frac{\partial^2 Y}{\partial\theta^2}\right\} + \frac{1}{\sin^2\theta Y}\frac{\partial^2 Y}{\partial\phi^2}$$

$\propto r$ $\propto \theta,\phi$

(introduce λ as the separation constant)

radial equation:

$$r^2(\mathcal{E} - V(r)) + \frac{\hbar^2}{2m}\left(\frac{r^2}{P}\frac{\partial^2 P}{\partial r^2}\right) = \lambda$$

$$\frac{\partial^2 P}{\partial r^2} + \frac{2m}{\hbar^2}\left(\mathcal{E} - V - \frac{\lambda}{r^2}\right)P(r) = 0$$

angular equation:

$$\lambda = \frac{1}{\sin\theta Y}\left\{\cos\theta\frac{\partial Y}{\partial\theta} + \sin\theta\frac{\partial^2 Y}{\partial\theta^2}\right\} + \frac{1}{\sin^2\theta Y}\frac{\partial^2 Y}{\partial\phi^2}$$

$$\Rightarrow \left[\frac{1}{\sin\theta}\left\{\cos\theta\frac{\partial}{\partial\theta} + \sin\theta\frac{\partial^2}{\partial\theta^2}\right\} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y = \lambda Y \quad \text{(e-val eqn)}$$

↓ if $\lambda = \ell(\ell+1)$ ← $\hbar^2$?

↓ solutions to this are spherical harmonics; $Y = Y_{\ell m}(\theta, \phi)$